GCF and LCM

The biggest factor they share. The smallest multiple they share.

Greatest common factor (GCF) of 12 and 18

Factors of 12: 1 2 3 4 6 12

Factors of 18: 1 2 3 6 9 18

1, 2, 3 and 6 are in both lists. The biggest one is 6.

GCF = 6

Least common multiple (LCM) of 4 and 6

Multiples of 4: 4 8 12 16 20 24

Multiples of 6: 6 12 18 24 30 36

Count by each number. 12 is the first one in both lists.

LCM = 12

Use the GCF to pull a sum apart

$$36 + 8 = 4 \times (9 + 2)$$

4 is the GCF of 36 and 8.
36 is 4 × 9 and 8 is 4 × 2.

Your turn!

GCF of 8 and 12 = ☐ LCM of 8 and 12 = ☐

GCF and LCM: Practice

Factors for the GCF. Multiples for the LCM.

Part 1 Common factors Ring the factors in both lists. The biggest one is the GCF.

1) Factors of 16: **1 2 4 8 16**

Factors of 24: **1 2 3 4 6 8 12 24**

Common factors are ringed. GCF = ☐

2) Factors of 15: **1 3 5 15**

Factors of 25: **1 5 25**

Common factors are ringed. GCF = ☐

3) Factors of 18: **1 2 3 6 9 18**

Factors of 27: **1 3 9 27**

Common factors are ringed. GCF = ☐

4) Factors of 20: **1 2 4 5 10 20**

Factors of 30: **1 2 3 5 6 10 15 30**

Common factors are ringed. GCF = ☐

5) Factors of 14: **1 2 7 14**

Factors of 21: **1 3 7 21**

Common factors are ringed. GCF = ☐

6) Factors of 12: **1 2 3 4 6 12**

Factors of 32: **1 2 4 8 16 32**

Common factors are ringed. GCF = ☐

GCF and LCM: Practice (continued)

Part 2 Find the GCF No help this time.

7) GCF of 24 and 36 = ☐

8) GCF of 40 and 56 = ☐

9) GCF of 27 and 45 = ☐

10) GCF of 32 and 48 = ☐

11) GCF of 35 and 63 = ☐

12) GCF of 42 and 70 = ☐

13) GCF of 36 and 54 = ☐

14) GCF of 26 and 39 = ☐

15) GCF of 48 and 80 = ☐

16) GCF of 21 and 49 = ☐

17) GCF of 30 and 75 = ☐

18) GCF of 44 and 66 = ☐

GCF and LCM: Practice (continued)

Part 2 Find the GCF (continued)

19) **GCF of 18 and 42 =** ☐

20) **GCF of 60 and 84 =** ☐

21) **GCF of 45 and 72 =** ☐

22) **GCF of 28 and 70 =** ☐

Part 3 Common multiples Ring the first number in both lists.

23)
Multiples of 3: **3 6 9 12 15 18 21 24**
Multiples of 4: **4 8 12 16 20 24 28 32**
The first one in both lists: LCM = ☐

24)
Multiples of 6: **6 12 18 24 30 36 42 48**
Multiples of 8: **8 16 24 32 40 48 56 64**
The first one in both lists: LCM = ☐

25)
Multiples of 4: **4 8 12 16 20 24 28 32**
Multiples of 10: **10 20 30 40 50 60 70 80**
The first one in both lists: LCM = ☐

26)
Multiples of 5: **5 10 15 20 25 30 35 40**
Multiples of 6: **6 12 18 24 30 36 42 48**
The first one in both lists: LCM = ☐

27)
Multiples of 9: **9 18 27 36 45 54 63 72**
Multiples of 12: **12 24 36 48 60 72 84 96**
The first one in both lists: LCM = ☐

GCF and LCM: Practice (continued)

Part 3 Common multiples (continued)

28) Multiples of 6: 6 12 18 24 30 36 42 48

Multiples of 10: 10 20 30 40 50 60 70 80

The first one in both lists: LCM = ☐

Part 4 Find the LCM No help this time.

29) **LCM of 4 and 6 = ☐**

30) **LCM of 3 and 8 = ☐**

31) **LCM of 6 and 9 = ☐**

32) **LCM of 10 and 12 = ☐**

33) **LCM of 5 and 7 = ☐**

34) **LCM of 8 and 12 = ☐**

35) **LCM of 4 and 9 = ☐**

36) **LCM of 6 and 7 = ☐**

37) **LCM of 2 and 9 = ☐**

38) **LCM of 9 and 10 = ☐**

GCF and LCM: Practice (continued)

Part 4 Find the LCM (continued)

39) LCM of 3 and 10 = ☐

40) LCM of 8 and 10 = ☐

41) LCM of 5 and 12 = ☐

42) LCM of 7 and 8 = ☐

43) LCM of 11 and 4 = ☐

44) LCM of 12 and 9 = ☐

Part 5 Pull out the GCF Write the GCF. The numbers in the ( ) have no common factor left.

45) $36 + 8 = \square \times (9 + 2)$

46) $45 + 30 = \square \times (3 + 2)$

47) $24 + 40 = \square \times (3 + 5)$

48) $14 + 35 = \square \times (2 + 5)$

49) $54 + 18 = \square \times (3 + 1)$

50) $27 + 63 = \square \times (3 + 7)$

GCF and LCM: Practice (continued)

Part 5 Pull out the GCF (continued)

51) $20 + 50 = \square \times (2 + 5)$

52) $32 + 56 = \square \times (4 + 7)$

53) $42 + 12 = \square \times (7 + 2)$

54) $64 + 40 = \square \times (8 + 5)$

Part 6 More GCF

55) GCF of 18 and 24 = ☐

56) GCF of 30 and 42 = ☐

57) GCF of 30 and 50 = ☐

58) GCF of 33 and 44 = ☐

59) GCF of 18 and 48 = ☐

60) GCF of 15 and 21 = ☐

61) GCF of 24 and 30 = ☐

62) GCF of 6 and 8 = ☐

GCF and LCM: Practice (continued)

Part 6 More GCF (continued)

63) GCF of 22 and 55 = ☐

64) GCF of 22 and 33 = ☐

65) GCF of 30 and 35 = ☐

66) GCF of 36 and 63 = ☐

67) GCF of 28 and 32 = ☐

68) GCF of 40 and 50 = ☐

69) GCF of 25 and 40 = ☐

70) GCF of 4 and 6 = ☐

71) GCF of 32 and 40 = ☐

72) GCF of 35 and 56 = ☐

GCF and LCM: Practice (continued)

Part 7 More LCM

73) LCM of 5 and 11 = ☐

74) LCM of 2 and 6 = ☐

75) LCM of 7 and 10 = ☐

76) LCM of 6 and 12 = ☐

77) LCM of 7 and 9 = ☐

78) LCM of 3 and 4 = ☐

79) LCM of 4 and 10 = ☐

80) LCM of 3 and 7 = ☐

81) LCM of 6 and 8 = ☐

82) LCM of 10 and 11 = ☐

83) LCM of 9 and 11 = ☐

84) LCM of 6 and 10 = ☐

GCF and LCM: Practice (continued)

Part 7 More LCM (continued)

85) LCM of 2 and 5 = ☐

86) LCM of 2 and 7 = ☐

Part 8 More pulling out the GCF

87) 72 + 63 = ☐ × (8 + 7)

88) 12 + 54 = ☐ × (2 + 9)

89) 21 + 6 = ☐ × (7 + 2)

90) 14 + 2 = ☐ × (7 + 1)

91) 36 + 81 = ☐ × (4 + 9)

92) 54 + 30 = ☐ × (9 + 5)

93) 18 + 16 = ☐ × (9 + 8)

94) 56 + 32 = ☐ × (7 + 4)

95) 81 + 9 = ☐ × (9 + 1)

96) 15 + 27 = ☐ × (5 + 9)

GCF and LCM: Practice (continued)

Part 9 Remember? Problems from earlier packets.

97) $2.6 \times 1.5 =$ ☐

98) 60% of 45 = ☐

99) $\frac{4}{7} \div \frac{1}{14} =$ ☐

100) 5,688 ÷ 24 = ☐

101) 8 : 5 = 32 : ☐

102) 3.92 ÷ 0.7 = ☐

Part 10 Word problems Factors or multiples? Read carefully.

103) Hot dogs come in packs of 8 and buns in packs of 6. What is the fewest of each so none are left?

Number sentence _____

Answer ☐ of each

Draw or show your work

104) 24 apples and 36 oranges go into the same number of bags, all alike. What is the most bags?

Number sentence _____

Answer ☐ bags

Draw or show your work

105) One bus comes every 10 minutes, another every 15. Both just came. In how many minutes again?

Number sentence _____

Answer ☐ minutes

Draw or show your work

GCF and LCM: Practice (continued)

Part 10 Word problems (continued)

106) A 42 cm by 56 cm board is cut into the biggest squares with no waste. How long is a side?

Number sentence _____

Answer ____ cm

Draw or show your work

107) Challenge: 18 boys and 27 girls make teams, all alike. What is the most teams? How many girls each?

Number sentence _____

Answer ____ girls

Draw or show your work

Check-In: GCF and LCM

Show what you know. No help this time. Score: _____ / 9

Part 1 Common factors

1) Factors of 18: 1 2 3 6 9 18

Factors of 30: 1 2 3 5 6 10 15 30

Common factors are ringed. GCF = ☐

Part 2 Find the GCF

2) **GCF of 33 and 55 = ☐**

3) **GCF of 36 and 60 = ☐**

4) **GCF of 28 and 42 = ☐**

Part 3 Find the LCM

5) **LCM of 12 and 8 = ☐**

6) **LCM of 6 and 15 = ☐**

7) **LCM of 9 and 6 = ☐**

Part 4 Pull out the GCF

8) **30 + 45 = ☐ × (2 + 3)**

9) **48 + 18 = ☐ × (8 + 3)**